Revision

2026-04-15

Overview

Plan for Day

We're going to do a couple more mixed strategy cases, then look over the sample exam.

Warm-Up Example 1: Penalty Kicks

A kicker and a goalie. The kicker wants to score, the goalie wants to save. The payoffs show goals scored (kicker's gain, goalie's loss), so this game is zero-sum.

	Goalie L	Goalie R
Kicker L	(0, 0)	(2, -2)
Kicker R	(2, -2)	(1, -1)

No Pure Nash Equilibrium

Row's best responses: vs. L, play R (2 > 0); vs. R, play L (2 > 1).

Column's best responses: vs. Kicker L, play L (0 > -2); vs. Kicker R, play R ($-1 > -2$).

The best responses cycle, so there is no pure-strategy NE.

Kicker's Mixing Probability

Let the kicker play L with probability p. Set the goalie indifferent.

Goalie L: $0 \cdot p + (-2)(1-p) = 2p - 2$

Goalie R: $(-2)p + (-1)(1-p) = -p - 1$

$$2p - 2 = -p - 1 \implies p = \frac{1}{3}$$

Goalie's Mixing Probability

Let the goalie play L with probability q. Set the kicker indifferent.

Kicker L: $0 \cdot q + 2(1-q) = 2 - 2q$

Kicker R: $2q + 1(1-q) = 1 + q$

$$2 - 2q = 1 + q \implies q = \frac{1}{3}$$

Kicker plays L with probability $\frac{1}{3}$. Goalie plays L with probability $\frac{1}{3}$.

Kicker's expected payoff: $2 - 2 \cdot \frac{1}{3} = \frac{4}{3}$.

Goalie's expected payoff: $-\frac{4}{3}$. The two payoffs sum to zero, as they must in a zero-sum game.

Warm-Up Example 2: Worker and Inspector

A worker chooses to shirk or work. An inspector chooses to inspect or rest.

Everyone gets a non-negative payoff, but incentives are opposed.

	Inspect	Rest
Shirk	(0, 2)	(4, 0)
Work	(2, 3)	(3, 4)

No Pure Nash Equilibrium

Row's best responses: vs. Inspect, play Work (2 > 0); vs. Rest, play Shirk (4 > 3).

Column's best responses: vs. Shirk, play Inspect (2 > 0); vs. Work, play Rest (4 > 3).

Again the best responses cycle, so there is no pure-strategy NE.

Let the worker play Shirk with probability p. Set the inspector indifferent.

Inspect: $2p + 3(1-p) = 3 - p$

Rest: $0 \cdot p + 4(1-p) = 4 - 4p$

$$3 - p = 4 - 4p \implies p = \frac{1}{3}$$

Let the inspector play Inspect with probability q. Set the worker indifferent.

$$\text{Shirk: } 0 \cdot q + 4(1-q) = 4 - 4q$$

$$\text{Work: } 2q + 3(1-q) = 3 - q$$

$$4 - 4q = 3 - q \implies q = \frac{1}{3}$$

Worker shirks with probability $\frac{1}{3}$. Inspector inspects with probability $\frac{1}{3}$.

Both players earn an expected payoff of $\frac{8}{3} \approx 2.67$.

The expected total is about 5.3. Since the game is positive-sum, the payoffs don't have to cancel, unlike the penalty-kick case.

The General 2x2 Case

The Setup

Any 2x2 game can be written in this general form:

	Left	Right
Up	(a, e)	(b, f)
Down	(c, g)	(d, h)

Suppose there is no pure-strategy NE. The same method works: each player mixes to make the other indifferent.

Finding p

Row plays Up with probability p. Column's expected payoffs:

$$\text{Left: } e \cdot p + g(1-p)$$

$$\text{Right: } f \cdot p + h(1-p)$$

Setting these equal and collecting terms in p:

$$p \cdot [(e+h) - (f+g)] = h - g$$

$$p = \frac{h-g}{(e+h)-(f+g)}$$

Finding q

Column plays Left with probability q. Row's expected payoffs:

$$\text{Up: } a \cdot q + b(1-q)$$

$$\text{Down: } c \cdot q + d(1-q)$$

By the same algebra:

$$q = \frac{d-b}{(a+d)-(b+c)}$$

The denominator for q is the sum of Row's main-diagonal payoffs minus the sum of Row's anti-diagonal payoffs. The denominator for p is the analogous quantity for Column.

If a denominator is zero, or if the solution for p or q lies outside $[0, 1]$, there is a pure-strategy NE and the mixed formula doesn't apply. This is rare in the textbook cases we'll see.

Question 1: Probability

The Setup

A payment processor flags potentially fraudulent transactions.

- 2% of transactions are fraudulent.
- The system correctly flags 90% of fraudulent transactions.
- It incorrectly flags 5% of legitimate transactions.

Part (a): P(flagged)

We need the law of total probability.

$$\Pr(G) = \Pr(G \mid F)\Pr(F) + \Pr(G \mid \neg F)\Pr(\neg F)$$

$$= (0.90)(0.02) + (0.05)(0.98)$$

$$= 0.018 + 0.049 = 0.067$$

About 6.7% of all transactions get flagged.

This is a Bayes' theorem question. We want $\Pr(F \mid G)$.

$$\Pr(F \mid G) = \frac{\Pr(G \mid F)\Pr(F)}{\Pr(G)} = \frac{0.018}{0.067} \approx 0.269$$

Only about 27% of flagged transactions are actually fraudulent. The base rate of fraud is so low that most flags are false positives, even with a pretty good detection rate.

Part (c): Independence

The question asks whether "flagged" and "fraudulent" are independent events.

Two events A and B are independent when $\mathsf{Pr}(A \mid B) = \mathsf{Pr}(A)$.

We have $\mathsf{Pr}(G \mid F) = 0.90$ and $\mathsf{Pr}(G) = 0.067$.

Since $0.90 \neq 0.067$, the events are not independent. Knowing a transaction is fraudulent changes the probability of it being flagged, which is the whole point of the system.

Question 2: Probability Trees

A student takes a two-part qualifying exam.

- $\Pr(\text{pass Part 1}) = 0.7$
- If pass Part 1: $\Pr(\text{pass Part 2}) = 0.8$
- If fail Part 1: $\Pr(\text{pass Part 2}) = 0.3$

$$\text{Pr}(P_1 \wedge P_2) = 0.7 \times 0.8 = 0.56$$

This is a direct application of the multiplication rule. The tree branches right to left.

Two ways to pass exactly one part:

- Pass Part 1, fail Part 2: $0.7 \times 0.2 = 0.14$
- Fail Part 1, pass Part 2: $0.3 \times 0.3 = 0.09$

Total: $0.14 + 0.09 = 0.23$.

Question 3: Expected Utility

The Setup

	Downturn (0.3)	Stable (0.4)	Growth (0.3)
Plan X	\$-\$90	80	100
Plan Y	10	30	50

Profits in thousands. The owner has \$100k in savings and utility $U = \sqrt{\text{wealth in thousands}}$.

$$\text{EMV}(X) = 0.3(-90) + 0.4(80) + 0.3(100) = -27 + 32 + 30 = 35$$

$$\text{EMV}(Y) = 0.3(10) + 0.4(30) + 0.3(50) = 3 + 12 + 15 = 30$$

Plan X has higher expected monetary value: \$35k vs. \$30k.

Part (b): Dominance

- Downturn: X gives -90, Y gives 10. Y is better.
- Stable: X gives 80, Y gives 30. X is better.

Each plan beats the other in at least one state. Neither dominates.

Part (c): Expected Utility — Setup

Total wealth = 100 + profit. Utility = $\sqrt{\text{wealth}}$.

Plan X wealth: 10, 180, 200.

Plan Y wealth: 110, 130, 150.

Notice the range: X's wealth goes from 10 to 200. Y's goes from 110 to 150. The square root function penalises low values, and X has a very low value in the Downturn scenario.

Part (c): The Calculation

$$EU(X) = 0.3\sqrt{10} + 0.4\sqrt{180} + 0.3\sqrt{200}$$

$$= 0.3(3.162) + 0.4(13.416) + 0.3(14.142) = 0.949 + 5.367 + 4.243 = 10.56$$

$$EU(Y) = 0.3\sqrt{110} + 0.4\sqrt{130} + 0.3\sqrt{150}$$

$$= 0.3(10.488) + 0.4(11.402) + 0.3(12.247) = 3.146 + 4.561 + 3.674 = 11.$$

Part (c): The Reversal

Plan X has higher EMV ($35k vs. $30k) but Plan Y has higher expected utility (11.38 vs. 10.56).

The concave utility function captures declining marginal utility of money. The owner cares proportionally more about avoiding the catastrophic Downturn outcome (wealth dropping to $10k) than about the upside in Growth.

This is the kind of case where EMV and EU come apart. The whole point of expected utility theory is that rational agents with curved utility functions should sometimes reject higher-EMV gambles.

Question 4: Strategic Form

	A	B	C	D
T	(4, 2)	(2, 1)	(3, 3)	(1, 0)
M	(2, 3)	(4, 0)	(1, 2)	(3, 1)
B	(1, 1)	(3, 2)	(2, 4)	(0, 0)

For Row, no strategy dominates another. Each pair has at least one reversal. For example, T beats M against A (4 vs. 2) but M beats T against B (4 vs. 2).

For Column, look at D's payoffs: 0, 1, 0.

Compare with A's payoffs: 2, 3, 1. Every entry is strictly higher. **A strictly dominates D.**

Compare with C's payoffs: 3, 2, 4. Every entry is strictly higher. **C strictly dominates D.**

Part (b): Iterated Elimination — Round 1

Eliminate D (strictly dominated). The reduced game:

	A	B	C
T	(4, 2)	(2, 1)	(3, 3)
M	(2, 3)	(4, 0)	(1, 2)
B	(1, 1)	(3, 2)	(2, 4)

Now look at Column's payoffs for B: 1, 0, 2. Compare with C: 3, 2, 4.

C beats B in every row (3 > 1, 2 > 0, 4 > 2). **C strictly dominates B.** Eliminate B.

	A	C
T	(4, 2)	(3, 3)
M	(2, 3)	(1, 2)
B	(1, 1)	(2, 4)

Now Row. Compare T with M: 4 > 2 against A, 3 > 1 against C. **T strictly dominates M.**

Compare T with B: 4 > 1 against A, 3 > 2 against C. **T strictly dominates B.**

Eliminate M and B. Column faces A (payoff 2) vs. C (payoff 3). Column picks C.

Solution: (T, C) with payoffs (3, 3).

Part (c): Nash Equilibria

Row's best responses (looking at Row's payoffs):

- vs. A: T gives 4. Best: **T**.
- vs. B: M gives 4. Best: **M**.
- vs. C: T gives 3. Best: **T**.
- vs. D: M gives 3. Best: **M**.

Column's best responses (looking at Column's payoffs):

- vs. T: C gives 3. Best: **C**.
- vs. M: A gives 3. Best: **A**.
- vs. B: C gives 4. Best: **C**.

The only cell where both players are playing best responses is (T, C).

T is Row's best response to C, and C is Column's best response to T.

Unique pure-strategy NE: (T, C) with payoffs (3, 3). This matches the iterated elimination result.

Question 5: Dynamic Games

The Game Tree

Player 1: A or B. Player 2 observes and responds.

- A $\longrightarrow$ P2 picks C or D. C gives (5, 3); D gives (2, 1).
- B $\longrightarrow$ P2 picks E or F. E gives (3, 4); F $\longrightarrow$ P1 picks G or H. G gives (6, 0); H gives (1, 5).

Part (a): Strategies

Player 1 moves at two nodes (root and after B $\longrightarrow$ F), so has 4 strategies: (A; G), (A; H), (B; G), (B; H).

Player 2 moves at two nodes (after A and after B), so has 4 strategies: (C; E), (C; F), (D; E), (D; F).

A strategy specifies what to do at every decision node, including ones that may never be reached. (A; G) and (A; H) are different strategies even though the G/H choice is moot when Player 1 plays A.

Part (b): Backward Induction — Step 1

Start at the deepest node: Player 1 after B $\longrightarrow$ F.

G gives Player 1 a payoff of 6. H gives 1. Player 1 chooses **G**.

Replace the node with (6, 0).

Part (b): Steps 2 and 3

Player 2's node after B: E gives P2 a payoff of 4; F leads to (6, 0), giving P2 payoff 0. Player 2 chooses **E**.

Player 2's node after A: C gives P2 payoff 3; D gives 1. Player 2 chooses **C**.

Player 1 at the root: A leads to (5, 3), giving P1 payoff 5. B leads to (3, 4), giving P1 payoff 3.

Player 1 chooses **A**.

Outcome: A $\rightarrow$ C, with payoffs (5, 3).

Player 2 prefers (3, 4) from B $\longrightarrow$ E. Player 2 might threaten: "If you play A, I'll play D, giving you only 2."

If believed, Player 1 would compare A $\longrightarrow$ D (payoff 2) with B $\longrightarrow$ E (payoff 3) and pick B.

But at P2's node after A: C gives P2 payoff 3, D gives 1. Player 2 won't follow through. The threat is not credible, and Player 1 knows it.

	(C; E)	(C; F)	(D; E)	(D; F)
(A; G)	(5, 3)	(5, 3)	(2, 1)	(2, 1)
(A; H)	(5, 3)	(5, 3)	(2, 1)	(2, 1)
(B; G)	(3, 4)	(6, 0)	(3, 4)	(6, 0)
(B; H)	(3, 4)	(1, 5)	(3, 4)	(1, 5)

Notice that the (A; G) and (A; H) rows are identical. When P1 chooses A, the G/H decision is never reached.

Also, the (C; E) and (D; E) columns differ only in the A-rows. When P2 chooses E after B, nothing else matters on that branch.

The backward-induction strategy profile is (A; G) for P1 and (C; E) for P2, yielding (5, 3). This is the subgame perfect equilibrium.

Question 6: Mixed Strategies

	Left	Right
Up	(3, 1)	(1, 2)
Down	(1, 3)	(2, 0)

No pure-strategy NE. (Check: Row's BR to Left is Up; Column's BR to Up is Right; Row's BR to Right is Down; Column's BR to Down is Left. The cycle means no cell has both players playing best responses.)

Let Row play Up with probability p. Set Column indifferent:

Left: $1 \cdot p + 3(1 - p) = 3 - 2p$

Right: $2p + 0(1 - p) = 2p$

$$3 - 2p = 2p \implies p = \frac{3}{4}$$

Let Column play Left with probability q. Set Row indifferent:

Up: $3q + 1(1 - q) = 2q + 1$

Down: $q + 2(1 - q) = 2 - q$

$$2q + 1 = 2 - q \implies q = \frac{1}{3}$$

The Equilibrium

Row plays Up with probability $\frac{3}{4}$, Down with probability $\frac{1}{4}$.

Column plays Left with probability $\frac{1}{3}$, Right with probability $\frac{2}{3}$.

Expected payoffs: Row gets $2 \times \frac{1}{3} + 1 = \frac{5}{3} \approx 1.67$. Column gets $3 - 2 \times \frac{3}{4} = \frac{3}{2} = 1.5$.

Both players earn less than their best pure-strategy payoff. Mixed equilibria are about making the opponent indifferent, not about maximising your own payoff.